

Supporting information

Species	Abundance	Length (mm)	Body Size	Feeding Trait	Wing Morphology
<i>Agonum (Olisares) impressum</i> (Panzer, 1796)	355	10.0	S	C	M
<i>Agonum (Olisares) viridicupreum</i> (Goeze, 1777)	34	10.0	S	C	M
<i>Amara curta</i> Dejean, 1828	1474	9.0	S	C	M
<i>Anchomenus dorsalis</i> (Pontoppidan, 1763)	605	8.0	S	O	M
<i>Anchomenus turkestanicus</i> Ballion, 1871	55	7.8	S	O	M
<i>Anisodactylus binotatus</i> (Fabricius, 1787)	495	13.0	S	C	B
<i>Callistus lunatus</i> Fabricius, 1775	201	7.0	S	O	M
<i>Carabus (Hemicarabus) tuberosus</i> Dejean, 1829	178	20.0	L	C	B
<i>Chlaenius (Chlaeniellus) nigricornis</i> Fabricius, 1787	163	12.0	S	C	M
<i>Harpalus (Harpalus) rubripes</i> (Dufischmid, 1812)	259	11.0	S	O	M
<i>Harpalus (Pseudoophonus) rufipes</i> (Degeer, 1774)	232	12.0	S	O	M
<i>Pterostichus (Adelosia) macer</i> (Marsham, 1802)	186	11.5	S	C	M
<i>Pterostichus (Argutor) vernalis</i> (Panzer, 1796)	117	11.0	S	C	M
<i>Poecilus (Poecilus) cupreus</i> (Linnaeus, 1758)	611	12.5	S	C	M
<i>Poecilus (Poecilus) punctulatus</i> (Schaller, 1783)	333	14.0	S	C	B
<i>Scarites terricola</i> Bonelli, 1813	155	21.0	L	C	B

Table S1: Identity, abundance, and functional traits of ground beetle individuals collected from 40 sampling plots in natural poplar forests along the Irtysh River in Xinjiang (Northwest China) (Notes. Ground beetles were classified as large (S, >15 mm) and small (L, <15 mm); carnivorous (C) and omnivorous (O); macropterous (M) and brachypterous (B)).

Low	Medium-	Medium	Medium-	High	SHDI	Canopy	Vegetation	SR of	Height	Dis. to
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	VP	low VP	VP	high VP	VP		cover	cover	herbs	of herbs	river
Low VP		0.78	-0.41	-0.14	0.20	-0.19	0.36	0.48	-0.71	-0.70	-0.23
Medium-low VP	0.78		-0.14	-0.04	-0.04	-0.04	0.28	0.27	-0.73	-0.74	-0.33
Medium VP	-0.41	-0.14		0.19	-0.63	0.63	-0.01	-0.07	0.19	0.18	-0.08
Medium-high VP	-0.14	-0.04	0.19		-0.63	0.08	-0.11	-0.22	-0.05	0.10	0.00
High VP	0.20	-0.04	-0.63	-0.63		-0.25	0.00	0.13	0.06	-0.11	0.34
SHDI	-0.19	-0.04	0.63	0.08	-0.25		-0.07	-0.03	0.13	0.02	0.02
Canopy cover	0.36	0.28	-0.01	-0.11	0.00	-0.07		0.31	-0.23	-0.25	-0.22
Vegetation cover	0.48	0.27	-0.07	-0.22	0.13	-0.03	0.31		-0.35	-0.31	-0.21
SR of herbs	-0.71	-0.73	0.19	-0.05	0.06	0.13	-0.23	-0.35		0.84	0.28
Height of herbs	-0.70	-0.74	0.18	0.10	-0.11	0.02	-0.25	-0.31	0.84		0.17
Dis. to river	-0.23	-0.33	-0.08	0.00	0.34	0.02	-0.22	-0.21	0.28	0.17	

Table S2: Spearman's correlation coefficients between landscape factors (proportions of low vegetation productivity (Low VP), medium-low vegetation productivity (Medium-low VP), medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), overall diversity of vegetation productivity classes (SHDI)) and stand characteristics (canopy cover, vegetation cover, species richness of herbaceous plant species (SR of herbs), average height of herbaceous plant species (Height of herbs) and the distance to river (Dis. to river)). Strong correlations (correlation coefficients $> |0.7|$) are highlighted in bold.

Response variables	Top-ranked models ($\Delta\text{AICc} < 2$)	df	AICc	ΔAICc	weight	R_m^2 (R_c^2)
Activity density	SHDI + Dis. to river	5	446.31	0.00	0.52	0.79(0.82)
	SHDI + Medium VP + Dis. to river	6	447.71	1.41	0.26	0.65(0.74)
	Vegetation cover + Medium VP + Canopy cover + Dis. to river	6	448.06	1.75	0.22	0.67(0.72)
Species diversity	SHDI + Medium VP + High VP + Vegetation cover + Dis. to river	9	385.50	0.00	0.55	0.76(0.80)
	SHDI + Medium VP + High VP + Canopy cover + Vegetation cover + Dis. to river	8	387.27	1.77	0.23	0.71(0.73)
	SHDI + Medium VP + Canopy cover + Vegetation cover + Dis. to river	9	387.35	1.85	0.22	0.64(0.67)
Functional diversity	Medium VP + Medium-high VP + High VP + Canopy cover + Vegetation cover + Dis. to river	9	435.29	0.00	0.59	0.73(0.79)
	Medium VP + Medium-high VP + High VP + Vegetation cover + Dis. to river	8	436.93	1.64	0.24	0.64(0.69)
	Medium VP + Canopy cover + Vegetation cover + Dis. to river	10	437.23	1.94	0.17	0.59(0.61)

Table S3: Summary of the top-ranked models for estimating responses of activity density, species diversity and functional diversity of ground beetles to the overall diversity of vegetation productivity classes (SHDI), the proportions of medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), canopy cover, vegetation cover and the nearest distance from sampling plot to river (Dis. to river) (Notes. The best-fit models are shown in bold; df, degree of freedom for each model; AICc, Akaike's Information Criterion corrected for small sample size; ΔAICc , difference in AICc value between the top-ranked models; Weight, Akaike weight, the conditional probabilities for each model; R_m^2 and R_c^2 , the proportion of variance explained by fixed effects and by both the fixed and random effects).